

Applied analytics & data systems — portfolio summary

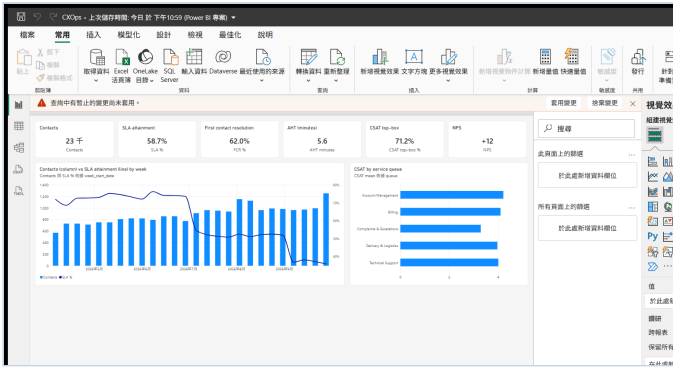
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1 · CUSTOMER-SERVICE OPERATIONS DASHBOARD — POWER BI, SQL, PYTHON

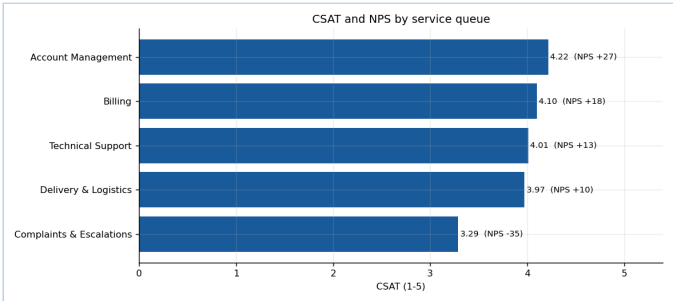
- Designed a **star-schema model** over 23,162 contacts and 15,721 cases (26 weeks, five channels), with a date table and six dimensions.
- Authored **36 DAX measures** covering SLA attainment, first-contact resolution, handle time, backlog ageing, CSAT/CES/NPS and cost per contact — each definition documented with its pitfalls.
- Added **30 automated data-quality checks** (key integrity, Likert ranges, flag coherence, calendar coverage) that gate a refresh instead of silently publishing wrong numbers.
- Every figure is validated against an **independent pandas implementation**, so the dashboard numbers are provably reproducible.



Finding	Evidence
Phone wait time doubled from week 14	CES 5.09 → 3.98 (−1.11, $p < .001$); CSAT 4.18 → 3.37
Email first-response SLA collapsed from week 23	80.6% → 23.9% ($p \approx 1e-296$); open backlog concentrated in 7–30 day bucket
Volume spike was <i>not</i> an experience problem	Billing contacts 153 → 302/week with CSAT flat (4.01 vs 4.11)

2 · CUSTOMER-EXPERIENCE METRICS & STATISTICAL ANALYSIS — SPSS, PYTHON

- Built an **SPSS-native dataset** (.sav with variable labels, value labels and measurement levels) plus full **SPSS syntax** for NPS/CES/CSAT, bootstrap confidence intervals, chi-square, Welch t-tests, ANOVA with Tukey post-hoc, Kruskal-Wallis, standardised-beta driver analysis and ordinal logistic regression (PLUM).
- NPS +11.6** (41.7% promoters, 30.1% detractors), bootstrap 95% CI [+9.1, +14.0] — resampled because NPS is a difference of proportions.
- First-contact resolution is the dominant driver of CSAT** (standardised $\beta = +0.49$, $R^2 = 0.347$), ahead of escalation (−0.24), transfers (−0.11) and wait time (−0.04).
- Channel and NPS group are **not independent** ($\chi^2 = 209.7$, $df = 8$, $p < .001$, Cramér's $V = 0.157$).
- Methodological control**: pooling wait time across channels suggests longer waits raise FCR (+0.39, $p < .001$); tested



within channel the effect disappears and reverses for live chat (-0.12 , $p = .002$) — a channel confound that would have produced the wrong recommendation.

Deliverables: 4-page bilingual findings report, SPSS `.sav` + `.sps`, and a procedure-by-procedure SPSS $\rightleftharpoons$ Python equivalence table. No SPSS licence was available on this machine, so the syntax is shipped with expected values and cross-checked numerically in Python — stated openly rather than faked.

HOW THIS MAPS TO THE ROLE

Requirement	Evidence in this portfolio
Develop and maintain databases and data systems	Star schema (7 tables), SQL data-quality gates, reproducible pipeline
Automate workflows for real-time analytics and performance tracking	Automated model/report generation from source files; 30 checks that block a bad refresh
Statistical tools for diagnosis and prediction	Bootstrap CIs, chi-square, Welch t-tests, ANOVA + Tukey, Kruskal-Wallis, regression, ordinal logistic
NPS, CES and satisfaction surveys	Full metric definitions, response-rate and non-response handling, driver analysis
Power BI, SPSS, Python, SQL, Excel/VBA	Used throughout; measure definitions, SPSS syntax and <code>.sav</code> files all included

Disclosure. All data is synthetic, generated by a seeded, documented generator; the projects demonstrate method and tooling, not any organisation's real performance. Every reported figure is reproducible by running the included scripts.

Repository: github.com/UNCHIKIT/cx-analytics-portfolio · Un Chi Kit · unchikit123@gmail.com · +853 6318 3022